

A 1.1.8 Compression



A 1.1.8 Describe the concept of compression

Every photo you send, every song you stream and every web page you load has almost certainly been compressed. Compression reduces the size of a file by representing the same information in fewer bits, which saves storage space and cuts transmission time. The syllabus asks you to describe the concept, to distinguish lossy methods from lossless methods, and to know two specific techniques: run-length encoding and transform coding. This page covers each in turn, with a worked example of run-length encoding and a conceptual account of transform coding.

Key Questions

- What is compression, and why is it used?
- How do lossy and lossless compression methods differ, and when is each the right choice?
- How do run-length encoding and transform coding reduce file size?

IB Command Term note

The command term here is **describe**: give a detailed account.

You are not asked to explain the mathematics behind any technique, but a description must include the defining property of each method. For lossless compression that property is exact reconstruction of the original; for lossy compression it is that some data is permanently discarded. Expect questions that ask you to distinguish the two methods, or to choose and justify a method for a given type of file. An answer that treats lossy and lossless as interchangeable cannot reach full marks.

What is compression?

Compression reduces file size by removing redundancy. Redundancy is information that is repeated, predictable or unnecessary for the purpose the file serves. A compression algorithm finds a shorter way of representing the same content, and a matching decompression algorithm rebuilds a usable file at the other end. The reasons we have compression are because smaller files take less storage, upload and download faster, and cost less to transmit, which is why compression sits underneath almost everything you do online.

A four-minute song stored as raw, uncompressed audio occupies roughly 40 MB. The same song as a 128 kbps MP3 file occupies about 4 MB. That tenfold reduction is the difference between streaming working smoothly on a phone connection and not working at all.

THINK FIRST

You need to shrink two files: a Python program and a holiday photo. For which file would losing a small amount of data be acceptable, and for which would it be a disaster?

Lossy and lossless: what is the difference?

Lossless compression removes only redundancy. Every bit of the original can be recovered, so decompressing the file gives a perfect copy. This is essential where we need to have an exact copy.

For example a program with one wrong byte will not run, and a spreadsheet with one wrong digit will give incorrect information. ZIP, gzip and PNG are examples of lossless files.

Lossy compression goes further. It permanently discards data judged unnecessary, usually detail that human eyes or ears barely register. The reward is a much higher compression ratio; the cost is that the original file can never be perfectly reconstructed. For photographs, music and video the trade is usually worth making, because a small, often imperceptible drop in quality brings a large saving in size. JPEG, MP3 and MPEG are lossy.

The choice between them is driven by the data, not by preference. To decide which one to use we ask one question: does every bit need to survive? If yes, the method must be lossless. If a close approximation will do, lossy methods deliver far smaller files.

Feature	Lossless compression	Lossy compression
Data loss	None	Some data permanently discarded
Reconstruction	Original recovered exactly	Original cannot be recovered

Feature	Lossless compression	Lossy compression
Compression ratio	Lower	Higher
Suited to	Text, program code, spreadsheets, data files	Images, audio, video
Examples	ZIP, gzip, PNG	JPEG, MP3, MPEG

COMMON EXAM MISTAKE

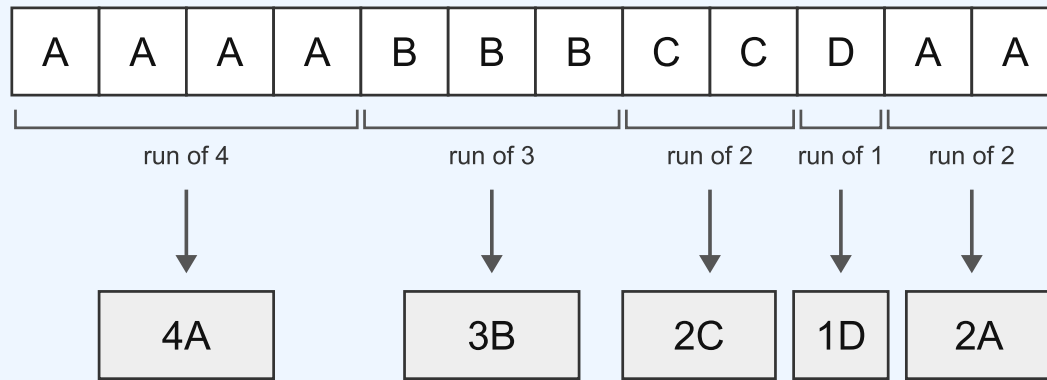
Students often define lossy compression as “lower quality” and lossless as “higher quality”. Quality is a consequence, not the definition. The key difference is about the reconstruction of the file: lossless methods allow the original file to be rebuilt exactly, while lossy methods discard data permanently, so the original cannot be rebuilt.

Run-length encoding

Run-length encoding (RLE) is a lossless technique that replaces a run (an unbroken sequence of the same value) with a single copy of the value and a count of how many times it repeats. Twelve characters of AAAABBBCCDAA become the ten characters 4A3B2C1D2A, and the decoder rebuilds the original exactly by expanding each pair.

RLE works well where long runs are common such as simple graphics with large blocks of one colour, fax transmissions, icons and screenshots. It fails where runs are rare. The string ABCD encodes as 1A1B1C1D, doubling in size, because every run has length one. A compression method can make a file larger when the data does not match its assumptions, which is itself a fact worth remembering for the exam.

Original string: 12 characters



Compressed string: 4A3B2C1D2A, 10 characters

Each run of repeated characters becomes a count followed by the character.

Figure 1. Run-length encoding compresses AAAABBBCCDAA (12 characters) to 4A3B2C1D2A (10 characters).

WORKED EXAMPLE



Transform coding

Transform coding is the idea behind lossy formats such as JPEG and MP3. Rather than storing raw values (the brightness of each pixel, the amplitude of each audio sample), it first converts the data into a different representation, the frequency domain, where the parts people barely perceive become easy to identify and remove.

HOW IT WORKS (CONCEPTUALLY)

Any image or sound can be described as a mixture of frequencies: slow, broad changes plus fast, fine detail. A mathematical transform rewrites the data in those terms. Human senses are sharp at perceiving the broad changes and poor at perceiving some of the fine ones, so the encoder deletes the components below the threshold of perception, then packs what remains using lossless techniques. The deletion is the lossy step: that data is gone for good, which is why a transform-coded file can never reproduce the original exactly. You only need to understand the concept, not the mathematics of the transform itself.

In images: JPEG

A JPEG encoder divides the image into small blocks of pixels and transforms each block into frequency components. Low frequencies carry the overall colour and brightness of the block; high frequencies carry sharp edges and fine texture. Because the eye tolerates losing fine texture far better than losing overall colour, the encoder rounds many high-frequency components down to zero. At sensible settings the loss is hard to see; pushed too far, it appears as the blocky, smeared artefacts familiar from heavily compressed images.

In audio: MP3

An MP3 encoder cuts the sound into short segments and transforms each into the frequencies it contains. It then removes what a listener cannot hear: sounds too quiet to register, and quieter sounds drowned out by a louder nearby frequency, an effect called masking. Discarding the inaudible content is why an MP3 file can be a tenth of the size of the raw recording while sounding nearly identical.

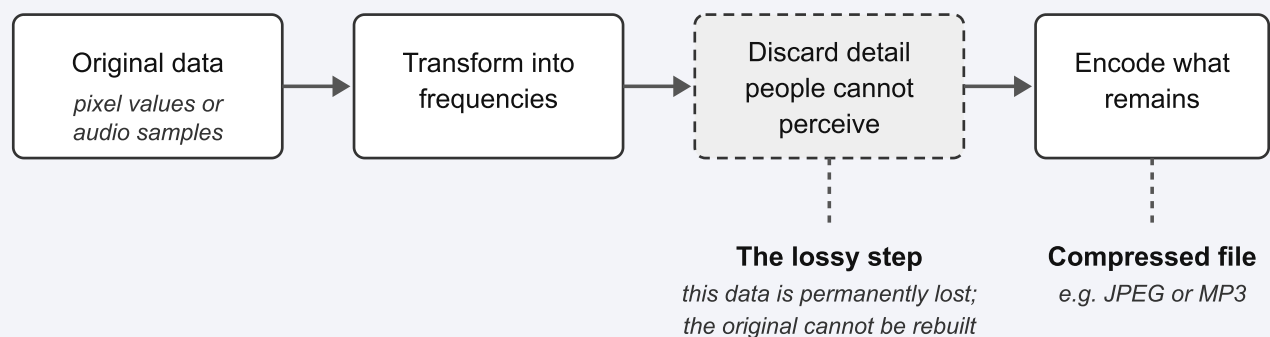


Figure 2. Transform coding: data is converted into frequencies, imperceptible detail is permanently discarded, and the remainder is encoded.

RLE and transform coding compared

The two named techniques sit on opposite sides of the lossy and lossless divide, which makes them a natural pairing for a compare-style question.

Feature	Run-length encoding	Transform coding
Category	Lossless	Lossy
Strategy	Replaces runs of repeated values with a count and the value	Converts data into frequencies, then discards imperceptible components

Feature	Run-length encoding	Transform coding
Representation	Stays in the original form of the data	Shifts into the frequency domain
Result	Original recovered exactly on decompression	Some data permanently lost; original cannot be recovered
Best suited to	Data with long runs, e.g. simple images with blocks of one colour	Photographs, music and video, e.g. JPEG and MP3

Summary

Concept	Key point
Compression	Reduces file size by removing redundancy, saving storage space and transmission time
Lossless compression	No data lost; the original file is reconstructed exactly; used for text, code and data (ZIP, gzip, PNG)
Lossy compression	Discards imperceptible data permanently for much smaller files; the original cannot be recovered (JPEG, MP3, MPEG)
Run-length encoding	Lossless; replaces each run of repeated values with a count and the value; effective only when runs are common
Transform coding	Lossy; converts data into the frequency domain, deletes components people cannot perceive, then encodes the rest

TRY IT ON PAPER

The quiz widget below has some multiple choice questions. These written questions help you practise the describe and apply skills needed in an exam. Answer on paper, then check against the model answers.

1. Outline one difference between lossy compression methods and lossless compression methods. [2 marks]

Model answer: Lossless compression allows the original file to be reconstructed exactly from the compressed file [1]. Lossy compression permanently discards some data, so the original file cannot be recovered [1].

2. Apply run-length encoding to the string AAAAAABBBBCCD. State the encoded string and the number of characters before and after encoding. [3 marks]

Model answer: The runs are AAAAAA, BBBB, CC and D [1]. The encoded string is 6A4B2C1D [1]. The original string has 13 characters and the encoded string has 8, so the encoding is shorter [1].

3. A student records a podcast and saves it as an MP3 file. Describe how transform coding reduces the size of the audio file. [4 marks]

Model answer: The audio is converted from raw samples into the frequencies it contains [1]. The encoder identifies sounds the listener cannot perceive, such as very quiet sounds or sounds masked by louder nearby frequencies [1]. These components are permanently discarded, which makes the method lossy [1]. The remaining data is encoded compactly, producing a much smaller file than the original recording [1].

A 1.1.8 Quiz

1

What is the primary goal of compression?

- A. ☐ To improve the quality of an image or audio file.
- B. ☐ To encrypt data for security.
- C. ☐ To convert data from one format to another.
- D. ☒ To reduce the size of a file for efficient storage and transfer.

2

Which type of compression allows for perfect reconstruction of the original file?

- A. ☐ Both lossy and lossless

B. ☐ Lossy

C. ☒ Lossless

D. ☐ Neither lossy nor lossless

3

Which of the following is a characteristic of lossy compression?

A. ☐ It is typically used for text files and programs.

B. ☐ It preserves all the original data.

C. ☒ It achieves higher compression ratios.

D. ☐ It is never used for images or audio

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Which technique is used in JPEG image compression to turn image data into frequency components?

A. ☐ Lempel-Ziv compression

B. ☒ Transform coding

C. ☐ Huffman coding

D. ☐ Run-length encoding

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What is the main idea behind run-length encoding (RLE)?

A. ☐ To encrypt data using a secret key.

- B. ☐ To assign shorter codes to more frequent data values.
- C. ☒ To replace repeating sequences of data with a single value and the number of times it repeats.
- D. ☐ To transform data into a different domain for efficient representation.

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Which of the following file types would typically use lossless compression?

- A. ☐ A movie file
- B. ☐ An MP3 music file
- C. ☒ A text document
- D. ☐ A high-resolution photograph

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What is a potential drawback of using lossy compression?

- A. ☒ The original file cannot be perfectly reconstructed.
- B. ☐ It is not compatible with any image or audio formats.
- C. ☐ It requires more processing power to compress and decompress.
- D. ☐ The compressed file might be larger than the original.

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Which compression technique is MOST suitable for compressing an image with large areas of the same colour?

- A. ☒ Run-length encoding

- B. ☐ Lempel-Ziv compression
- C. ☐ Huffman coding
- D. ☐ Transform coding

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What is a common misconception about compression?

- A. ☒ All files can be compressed to any desired size.
- B. ☐ Compression is a relatively new technology
- C. ☐ Lossy compression always results in noticeable quality loss.
- D. ☐ Compression is only used for storing files, not for transmitting them.

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Why is it important to understand the different types of compression?

- A. ☐ It is not important for most computer users.
- B. ☐ To impress friends with your knowledge of file formats.
- C. ☒ To choose the most appropriate compression method for different types of data and apply it.
- D. ☐ To be able to write your own compression algorithms.



